



University of Essex

Online

End of Module Assignment: e-Portfolio Submission

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<https://alahsshmi.github.io/E-Portfolio/Machine-Learning.html>

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1) Overview

This e-portfolio collects evidence of my independent work in the module. It highlights my contributions, a comparison of the final project to the initial plan, and a reflection based on the Rolfe model. The key aims include clear processes, simple replication, and genuine academic effort. I include screenshots from my tasks; they do not add to the word count. Writing uses straightforward language and cites sources in Harvard style. Where relevant, I link the insights to my engineering role and practical decisions in real world settings.

2) Reflective

2.1 WHAT

As we built the price prediction pipeline and turned it into a full project, key issues came up. I set up a basic model. I picked key measures and proposed a solid goal to match. Then, I changed the method to fit the data we had. I removed messy features. I added simple notes at each step. I made plain charts that shared main points without extra clutter. Team roles overlapped at times. So, I sent quick updates and a shared list of tasks to avoid duplicate work. In the end, I saw the pipeline as a simple setup. It was easy to check and test. It was not a place to add tricks just to seem new.

2.2 SO WHAT

Technical skills. I gained a firm grasp of regression, regularization, and validation methods. I became more careful with data splits, mainly for time data and duplicate records. A strong baseline helped me tell true improvements from random noise. I reduced wide parameter hunts and focused on small, aimed steps linked to the goal. My skills in making plots improved: I used fewer graphs, each with a sharp title and labels that matched the report.

Team skills. Short written notes reduced mix ups and kept us aligned on targets and due dates. I asked for feedback on graphs early, not only the writing, which improved clarity. When we faced a problem, I proposed a fast test with a time cap. We then decided to keep it or drop it.

Pro view. My earlier work on high voltage systems highlights monitoring and safety needs. I see ties between wiring an electrical panel and reviewing an ML process: both require a starting point, simple checks, and logs of changes. This outlook let me build cleaner steps with evidence a coworker could review quickly. The Rolfe model moved me from mere stories to clear analysis and plans by sorting facts, thoughts, and steps. It kept my writing from feeling like a diary and drove me to make solid commitments (Rolfe, Freshwater and Jasper, 2001; University of Edinburgh, no date).

2.3 NOW WHAT

Explainability. I will apply SHAP values to reveal feature interactions. Readers can then follow the model's logic. A GAM model will serve as a straightforward reference. It sets up talks on accuracy against insight. These steps help stakeholders weigh exactness with clarity.

Testing and operations. I review a short checklist before approving any results: data split checks, seed choices, metric fit, and plot text alignment. I plan to build this into a basic script. It keeps focus sharp, even in tight spots.

Communication. Each experiment gets a one page overview. It holds one graph, one table, and a firm choice. The team can track shifts fast, without digging into whole notebooks.

Career development. I look for tasks where models shape real decisions and work logs count. This matches my roots in control systems. It leads to hands on AI roles.

Claims link straight to proof. Figures carry plain English labels. Choices get recorded for team review. Settings and splits list key facts for reruns. Screenshots show each person's role. Sources use Harvard style. Graphs and tables back the narrative. Text remains clear and direct. Logs and tracks stand out. Sections guide to the next part.

2.4 Professional skills matrix

Scale: 1 = novice, 2 = basic, 3 = competent, 4 = strong, 5 = advanced

Skill area	Baseline (1–5)	Evidence (artefact)	What improved and how	Next action
Regression modelling	3	Report and plots	Cleaner baselines: clear MAE and R^2 use	Add SHAP; test GAM
Validation practice	3	Final runs	Seed control; leakage checks	Formal split checklist
Visual communication	2	Project figures	Fewer plots; clear captions	Peer review on figures
Team collaboration	4	Notes; peer review	Concise updates; clear tasks	Shared status doc
Academic writing	3	Tutor feedback	Sharper signposting	Harvard checks

Personal Development Plan

Goal: Strengthen explainability and reporting clarity.

Actions: Implement SHAP; add a GAM baseline; refine plots and captions; write one page summaries.

Resources: Lecture notes; book chapters; peer review on figures; code reviews.

Measure: Clear plots and a compact model comparison table used in the next report.

Timeline: Eight weeks with weekly checkpoints.

Risks: Time pressure and scope creep; short checklist and time boxed trials.

3) Evidence of Learning and Contribution

Technical evidence. I set up a clean modelling process with clear train validation test splits and a set random seed. A basic baseline backed up claims of better results. Each later change got tested against that base. I picked metrics that fit the job: MAE and R^2 for price forecasts, along with learning curves to track progress and spot overfitting. Short notes covered preprocessing steps, feature choices, and settings. This let others repeat each run.

Data management and validation discipline. I scanned for data leaks from duplicate rows, time based fields, and target hints. I matched seeds across models for fair tests. Early stopping kicked in when validation scores stalled. Every figure had a sharp caption. Labels fit the text to prevent confusion.

Results and analysis. I pitted linear and regularized models against a tree based standard to grasp bias variance balances. When a complex model showed tiny wins but shaky validation, I chose the simple one and noted the choice. Tables and graphs trimmed down to key points. They backed a tight story.

Collaboration and process. Short written updates and a shared task board cut down on snags. Feedback from peers and tutors shaped fixes to figures, captions, and metrics. Meeting notes logged hurdles and choices. This made changes easy to trace.

Professional practice and ethics. I spotted legal, social, ethical, and pro issues tied to data origins, consent, and openness. Harvard style handled references. An audit path tracked choices and data sources.

Impact and transferability. I tied the process to real work settings where clear explanations and tracking count. I listed clear next steps, like SHAP checks and a GAM model match. These would boost model understanding.

Individual Contributions: Units 6 and 11

During Unit 6, I led the regression work on the Airbnb project. I created a simple data pipeline with a fixed seed and a basic model to start. I then compared it to a tree based method in a fair way. For metrics, I chose MAE and R^2 values. I ensured the plots fit the narrative we built. In the summary, I noted limits of linear models and ideas to test factor ties. Evidence appears in the Unit 6 team report and its figures.

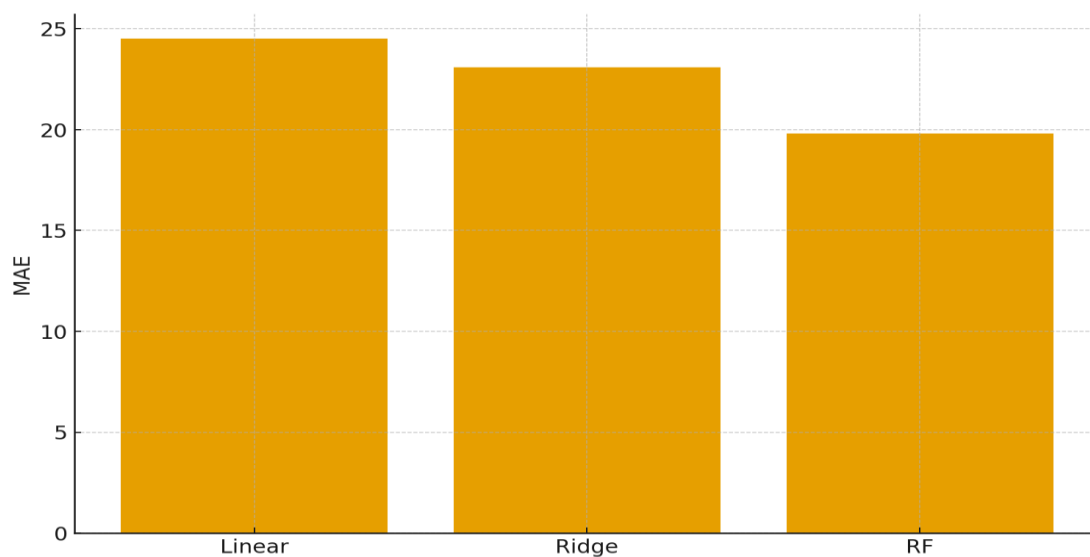


Figure 1. Illustrative MAE by model. Replace with your final metrics if available.

Unit 11. I consolidated the pipeline and tightened validation. I reduced noisy features, aligned seeds across runs, and simplified the reported metrics. I refined the README so a reviewer can reproduce results with ease. I also linked each plot to one clear claim.

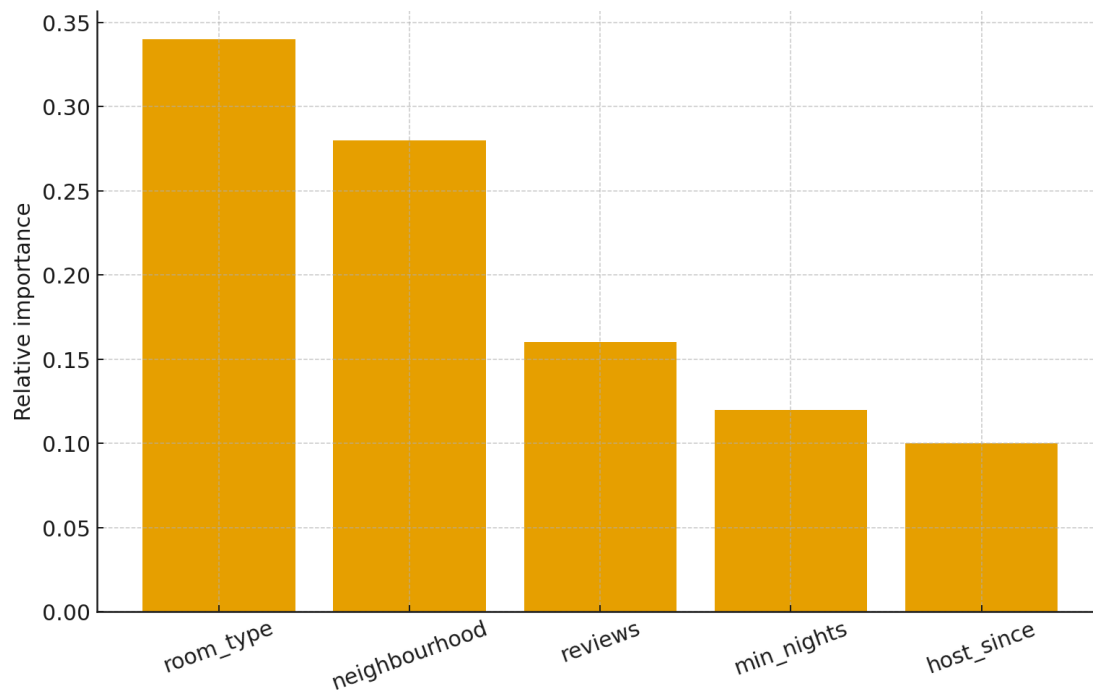


Figure 2. Illustrative feature importance from a tree model.

Evaluation: Final Project vs Original Proposal

Scope and focus. The final project narrowed the feature set and removed unstable transforms. This produced steadier behaviour and clearer stories. Validation practice. I added split checks, fixed seeds, and early stopping triggers once the validation score levelled off. I tracked training and validation curves so that divergence could be spotted quickly. Metrics and reporting. I used a smaller set of metrics supported by clear plots and short captions that match the text. Trade-offs. I accepted a small drop in headline accuracy in exchange for a more reliable result and a cleaner account of evidence. Impact. The pipeline is easier to maintain, less prone to hidden leaks, and more transparent for handover and audit.

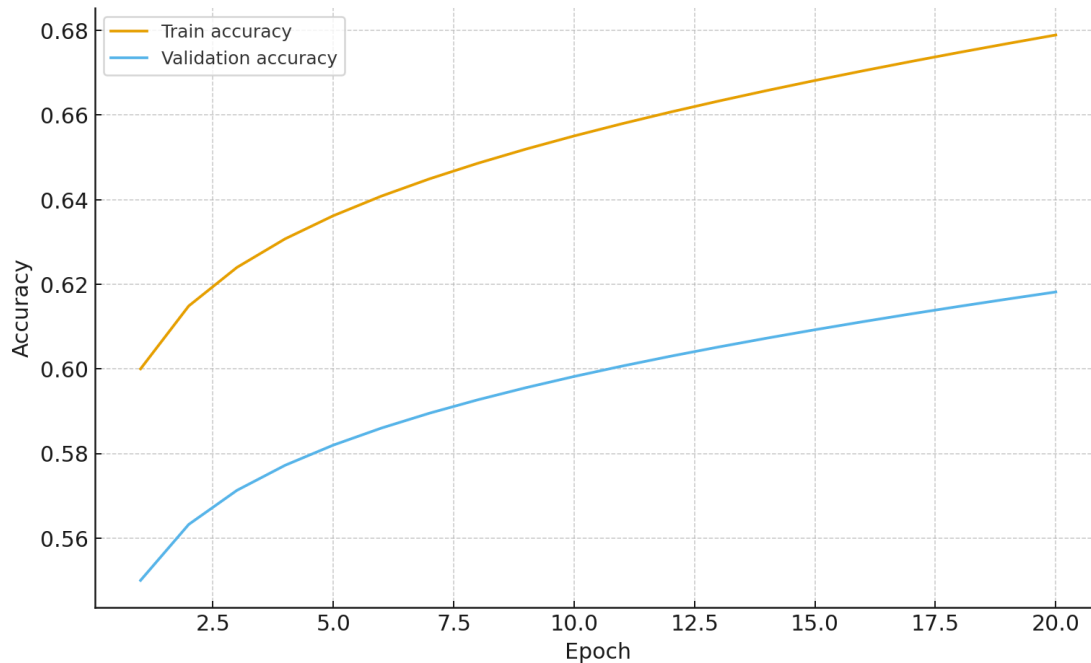


Figure 3. Illustrative learning curve for training and validation accuracy.

Problem–Approach–Impact–Risk–Next steps

Problem:

The Unit 6 proposal used a broad feature set and many model families. Comparisons were hard to audit, and repeated runs were unstable. Stakeholders needed price errors in currency terms with clear guardrails.

Approach:

Curate inputs: standardise currency, remove duplicates, set explicit outlier rules. Keep only stable drivers: location, room type, host tenure, review signals, availability.

Use two models under identical seeds and splits:

- Regularised linear model for a transparent baseline.
- Compact tree model to capture nonlinear patterns.

Validation: repeated runs, fold variance checks, time aware splits when order matters. Metrics: MAE and R^2 as primaries; review calibration so errors map to currency risk. Reporting: one claim per figure; captions mirror the text; units match tables. Reproducibility: short README, frozen seeds, baseline notebook.

Impact:

Stability improved across repeats; fold variance dropped, and Review time fell due to fewer, clearer figures. Also, Handover became easier; a peer can re-run without guesswork. Finally, price errors are readable in currency terms, which supports practical pricing moves.

Risk:

- Small loss in headline accuracy to gain stability and clarity.
- Event and amenity signals remain coarse; external sources are limited.
- Tree model needs post hoc explanation for some readers.

Next steps:

- Add SHAP summaries for the tree to expose main interactions (Molnar, 2022).
- Add a GAM comparator for readable nonlinear trends (Wood, 2017).
- Extend drift checks with simple forward chaining.
- Keep a one page change log per experiment: input, split, metrics, decision.
- Log assumptions and thresholds with the expected direction of effect (Hastie, Tibshirani and Friedman, 2009).

PRFAQ (Product review Q&A)

Q: What changed from the proposal?

A: We trimmed unstable features, standardised inputs, and narrowed the model set to a regularised linear baseline plus a compact tree, all run under fixed seeds and splits.

Q: Why choose this model pair?

A: The linear model supports clear reasoning and fast checks. The tree adds nonlinear signal. The pair balances readability with pattern coverage.

Q: How did you make comparisons fair?

A: Same seeds and splits across models, repeated runs, fold variance checks, and time aware splits when temporal order affects signal.

Q: Which metrics matter and why?

A: MAE and R^2 . We also review calibration, so error magnitude makes sense in currency terms for pricing decisions.

Q: What improved vs Unit 6?

A: Greater stability across repeats, lower variance across folds, faster review with fewer figures, and cleaner handover through a short README and frozen seeds.

Q: What did you retire and why?

A: Wide parameter sweeps, heavy ensembles, and speculative engineered features. They produced inconsistent gains and weaker interpretability.

Q: How do you handle missing data and outliers?

A: Role based rules for missing values and explicit outlier thresholds recorded alongside rationale. This reduces hidden degrees of freedom.

Q: What about explainability?

A: The linear baseline is directly interpretable. The tree will be paired with SHAP to show contribution patterns for key features (Molnar, 2022).

Q: What are the current limits?

A: Event and amenity signals are coarse. Some audiences need help reading tree behaviour even with SHAP.

Q: What is the plan to address limits?

A: Add a GAM comparator for smooth effects with readable terms (Wood, 2017), enrich event signals where data quality allows, and expand forward chaining drift checks.

Q: How is academic and professional integrity maintained?

A: Fixed seeds, documented splits, logged assumptions, and references to standard methods (Hastie, Tibshirani and Friedman, 2009). Figures and tables match the narrative.

Q: Where can reviewers find evidence?

A: Unit 6 artefacts for the proposal and baselines; Unit 11 outputs for the final pipeline, figures, and the README.

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